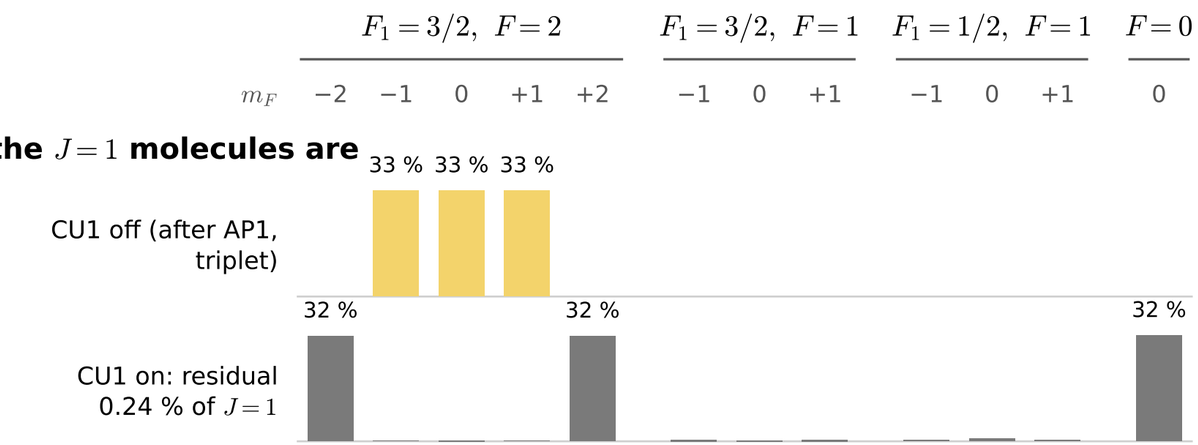
What each $J=1$ detection line tells about CU1



(b) photons per molecule from each sublevel

(c) predicted $S^{\mathrm{on}}/S^{\mathrm{off}}$, and what it tells about CU1

	$F_1=3/2,\ F=2$					$F_1=3/2,\ F=1$			$F_1=1/2,\ F=1$			$F=0$		
	−2	−1	0	+1	+2	−1	0	+1	−1	0	+1	0		
R(1) $F_1'=5/2,\ F'=3$ any polarization	2.1	2.1	2.1	2.1	2.1	0	0	0	0	0	0	0	**0.15 %**	$F=2$ only, no dark states → the $F=2$ part of the residual, exactly. Blind to $F=0$ (32 % of the residual) and to the singlet.
Q(1) $F_1'=1/2,\ F'=1$ polarization ∥ axis	0	2.9	4.4	2.9	0	2.5	0	2.5	3.1	0	3.1	4.4	**0.10 %**	$F=0$ is never dark (4.4), but $F=2,\ m_F=\pm2$ is: misses 64 % of the residual. Gives an upper bound on the $F=0$ part.
Q(1) $F_1'=1/2,\ F'=1$ polarization ⊥ axis	3.1	1.3	0.9	1.3	3.1	1.1	2.9	1.1	1.4	3.7	1.4	3.7	**0.64 %**	Sees all 12, but with 0.9–3.7 photons: the CU1-off levels weigh little, the residual a lot → 2.7× too high (approx.).

Both lines together

$$\frac{N^{\mathrm{on}}_{J=1}}{N^{\mathrm{off}}_{J=1}} \lesssim \frac{S^{\mathrm{on}}_R/2.1 \;+\; S^{\mathrm{on}}_Q/4.4}{S^{\mathrm{off}}_R/2.1}$$

= 0.23–0.33 % for any alignment (true 0.24 %)

S_R, S_Q: signals on R(1) $F'=3$ and Q(1) $F'=1$; 2.1, 4.4: photons per $F=2$ molecule (R) and per $F=0$ molecule (Q). $F=2$ exact, $F=0$ bounded, the $F=1$ levels (3 %) from the model.

Q(1) at ∥ and ⊥: $S^{\mathrm{on}}_Q/S^{\mathrm{off}}_Q$ = 0.10 % vs 0.64 % if the molecules keep the CU1 field axis, the same if their alignment is scrambled → tests the alignment and the CU1 model.

Readings predicted for CU1 at +19.0 MHz, 4.8 cm beam, on axis, AP1 triplet start (true $J=1$ residual 0.24 %); probe saturated (50 mW/cm², centre, $v_x=0$), zero field.
∥ / ⊥: probe polarization relative to the molecules' quantization axis; ⊥ approximate (no decay-induced ground coherence in the solver).